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How dominant digital technologies in schools influence the enactment of curriculum change

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ABSTRACT

This paper focuses on a large-scale curriculum change in second level education in Ireland to show how the dominance of existing technologies in schools, in this case slideware technologies, can influence the enactment of large-scale curriculum change. Through a conceptual review of the existing literature published in relation to the roll-out of the curriculum changes and digital technology use in schools, the paper highlights how slideware technologies can influence the ways in which students are encouraged to present and evidence their learning as part of mandated project work in schools. Furthermore, the paper contends that an emphasis on such software tools, while aiming to prepare students with digital skills for the future workplace, also inculcates the student into a corporate culture with its related neoliberal and consumerist ideologies. What could be seen as an innocuous piece of software in schools, can have a significant backwash effect on how curriculum change is enacted by teachers and experienced by learners in classrooms. The paper concludes by considering what is also locked-out through the dominance of such software.

IMPACT STATEMENT

This paper shows how dominant, taken-for-granted digital tools in schools can influence the enactment of curriculum reforms. Drawing on a case of curriculum change at lower-secondary education in Ireland, the paper argues that the widespread use and availability of slideware technology (e.g., PowerPoint) can influence how Classroom-Based Assessments (CBAs) are understood, produced, and presented. The study highlights the "backwash" effect of this dominance, including pressures toward linear, bullet-pointed representations of learning and an instrumental focus on "workplace-ready" digital skills. The paper's contribution is to make visible what is often overlooked: curriculum reforms are mediated by existing technological infrastructures, which can both enable change and lock-out alternative forms of expression and learning.

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Introduction

In a recent exploration of the use of Chromebooks in Danish classrooms, Cone and Lai (2025) argue that the introduction of technology platforms into schools, and education systems at a macro-level can 'lock-in' education systems into particular modes of working and constrain their options to adopt other technologies. They identified five types of 'lock-in' (data, political, regulatory, discursive, and temporal) that can occur as a result of schools developing an infrastructural dependency on specific technologies. Data lock-in occurs when schools become dependent on the data infrastructures of the platform provider, limiting options to migrate data elsewhere. Political lock-in arises because digital technologies become aligned with broader policy agendas that can be difficult to reverse regardless of the problems that can arise. Regulatory lock-in reflects the gap between legislation and practice where it can be difficult to enforce rules around data and privacy compliance with large technology companies. Discursive lock-in

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refers to the limiting of public debate around binary choices that centre on technology use limiting the possibility of other. Temporal lock-in refers to the growing dependency on specific platforms the longer they are used. These different dimensions of 'lock-in' reduce choice for schools and increasingly limit autonomy. While situated in the Danish context, the types of lock-in described are reflected in international education systems that are adopting platforms and the wider digital ecosystems that they provide. With this in mind, this paper turns to look specifically at the Irish context and in particular a recent large-scale curriculum change in second-level education. While Irish schools are not mandated to adopt particular software platforms and instead have a high level of autonomy in selecting digital resources to support their work, corporations such as Google, Microsoft, Apple and other EdTech providers have a significant footprint. These companies, often through third party vendors, offer a full suite of services to schools that often lack the technical expertise to adopt digital technologies independently. These schools can therefore become locked-in and reliant on their specific technologies and products. Through a conceptual review of the available literature and drawing on multiple sources of published work from the Irish context, this paper provides evidence of how large-scale curriculum change can be impacted by the digital technologies that schools traditionally use and how new pedagogical practices mandated by curriculum reforms can be mediated by existing technologies. While Cone and Lai (2025) focus on the macro level and impact, this paper considers lock-in from the classroom perspective – what impact does lock-in have on teacher and learners practice in the classroom? The unique contribution of this work is in highlighting how the adoption of specific platforms not only lock-in school in the ways outlined by Cone and Lai (2025), but this 'locking-in' extends to the classroom level. The dominance of digital platforms can influence the enactment of curriculum reforms and determine how key features of the curriculum are experienced by teachers and students.

To achieve this, the paper explores the enactment of Classroom Based Assessments (CBAs), a type of project-based learning introduced as part of curriculum changes to lower second level schooling in Ireland, namely the Framework for Junior Cycle and examines the extent to which the traditional dominant use of slideware technologies in Irish second-level classrooms influenced how these CBAs were enacted. Slideware technologies refer to software tools that create and display slide-based presentations. They combine text, images, charts, video/audio on slides that are presented in a linear sequence. Microsoft PowerPoint, Google Slides and Apple Keynote are commonly used examples. Guided by the research question, 'what evidence exists in the literature that schools have become locked-in to using slideware technologies in enacting the CBAs?', the paper highlights how a dependency on particular software products can significantly influence how broader curriculum changes are enacted and ultimately realised at a classroom level. In doing so, we hope to encourage a critical reflection on the taken-for-granted technology-mediated practices that schools engage with on a daily basis and encourage consideration of the backwash effect and unintended consequences reliance on such software products can have on the nature of teaching and learning.

The paper firstly provides information on the curriculum change context in Ireland and the role that digital technologies have traditionally played in classrooms in Irish second level schools. Following this, we offer a review of the current literature in relation to digital technology use and CBAs providing evidence of how slideware technologies have become the dominant mode of presenting student project work at lower second level education. To do so we draw on various studies that have been conducted over the past decade in Ireland. The final section explores some potential ways in which such software can lock-in students to ways of thinking and working in relation to their project work and discusses the implications of this.

Curriculum change at lower second-level education in Ireland

Curriculum change in Ireland is undertaken through a partnership model involving all stakeholders in education. The National Council for Curriculum and Assessment (NCCA), a statutory body under the auspices of the Department of Education and Youth, is responsible for the revision of existing, and development of new curricula. When devising new curricula or revising existing curricula, the NCCA convenes development groups consisting of representatives from the various stakeholder groups including teachers, parents, educational management bodies, industry and other partners. These groups work on

devising new curricula, mindful of existing practices and new thinking. It is often informed by research undertaken to inform the work of these committees. Following this model, the old Junior Certificate, lower second level (aged 12-15), was revised. Commencing in 2010 the consultation document, *Innovations and Identity* (NCCA, 2011), set out a number of limitations of the existing Junior Certificate. These included the excessive emphasis on examinations (and the backwash effect this had on the nature of teaching and learning in classrooms), the disengagement of students as a result of the acquisition of content they did not deem as necessary for their lives, the passive approaches to teaching and learning evident to many subjects, the lack of emphasis on the importance of acquiring skills for the 21st century and an over emphasis on the terminal examinations. This report was also informed by earlier studies (Smyth et al. 2006; Smyth & Hannan, 2007) exploring students' experiences of the Junior Certificate. Smyth et al. (2006) and Smyth and Hannan (2007) work highlighted that students were largely disengaged from learning, particularly specific cohorts of students and that the emphasis on terminal examinations created a high level of anxiety amongst some students.

The 'Innovations and Identity' report (NCCA, 2010) set out a vision for lower second-level education arguing that it needed to be decoupled from upper second-level education (the Leaving Certificate) which was characterised by a high stakes terminal examination that determined entry into university education. Because of this, the Leaving Certificate had a very strong emphasis on the regurgitation of information for examinations, and this influenced the nature of teaching and learning in the Junior Certificate, which was seen as a 'trial run' for the Leaving Certificate. In setting out this new vision, it aimed to fundamentally change the relationship between teaching, learning and assessment. It aimed to reduce the emphasis on terminal examination and create a greater balance between knowledge and skills, which at the time had an excessive emphasis on knowledge acquisition. The Framework for Junior Cycle was introduced in 2012. Its initial introduction included changes that went beyond what was set out and agreed through the partnership and consultation processes of the NCCA. As a result of this, there was significant resistance from teacher unions to the changes, leading to teacher strikes and a negotiated settlement in 2015 where many aspects of the change were modified. Over the following years, these curriculum changes were introduced on a phased basis with the final set of revised school subjects being introduced in 2020 (MacPhail et al., 2018).

A significant change to all subjects was the introduction of Classroom-based Assessments (CBAs). CBAs mandated all students to undertake a form of project-based learning in all subject areas. These CBAs were introduced to provide greater choice and flexibility for both teachers and students, to increase student engagement and motivation and to facilitate the development of important 21st century skills that the Framework for Junior Cycle aimed to achieve. In some subjects that had a traditional vocational emphasis, the introduction of project-based learning was not seen as a particularly novel or challenging step (McGarr et al., 2024). However, in other more traditional subjects characterised by strong emphasis on content and passive student learning, the introduction of project-based learning was a significant change. In many of the CBAs across different subjects', students were provided with the opportunity to undertake a project/investigation on a topic of their choosing culminating in the completion of a project report and presentation (For examples of CBA guidelines see: Microsoft Word - Assessment guidelines_Science_March_2018_For Principles.docx & JCPE_135_AssessmentGuidelines_EN.pdf.

Digital technologies in the Irish education system

Before looking at the role technology played in the realisation of the CBAs, it is firstly important to outline the role of digital technologies in the Irish post-primary education system.

International and national studies have found that educational policy and discourse have tended to position digital technology as modern and progressive (McGarr & Johnston, 2021), as providing the solution to all and any educational ills (Suárez-Guerrero et al., 2023), and as possessing magical powers (Ryan et al., 2020). Integrating and using digital technology is frequently presented as something that is unquestionable, inevitable and something modern and effective schools do (Ryan et al., 2020). Ireland is no exception to this, as there has been a climate of techno-positivity in education for many decades. Efforts to integrate digital technologies in teaching and learning in Irish post-primary schools have been

a feature of educational policy since the late 1990s (McGarr & Johnston, 2021). Influenced by policy initiatives across Europe and other western countries in the late 1990's, the Irish government introduced the *Schools IT 2000* project which was the first large-scale policy initiative aimed at increasing the use of digital technologies in Irish classrooms (DES, 1997). This policy was followed by a series of similar initiatives throughout the following two decades, all aiming to increase digital technology use across the curriculum and equip students with the necessary digital technology skills (2000, 2008b, 2009, 2015, 2017, 2022). This strong policy agenda, coupled with the presence of many global digital corporations within the Irish economy, has sustained a techno-positive climate relating to digital technologies in education. Techno-positive discourses around the benefits of digital technologies and teaching and learning and the critical importance of digital technology skills to sustain competitive digital economy in a global world are not contested within the Irish context (McGarr, 2023). Recent years have also seen the introduction of other curriculum innovations in the area of digital technologies including the introduction of short courses at lower second level education in the area of digital media and coding as well as the introduction of a Leaving Certificate Computer Science subject at upper second level education (Borges et al., 2024).

Successive studies into the nature and use of digital technologies in schools however have indicated that they tend to have been used to support existing modes of teaching and learning in classrooms rather than having a disruptive effect. For example, work by departmental inspectors showed an excessive emphasis on the use of slideware technologies in teacher presentations to their students (DES, 2008a). Similarly, other work has also indicated the limited extent to which digital technologies have been fully integrated across the curriculum. With the exception of curriculum successes in the area of coding at lower second level education and computer science at upper second level education, the extent to which digital technologies have been implemented across the curriculum have been tempered by both resource constraints and an emphasis on examination preparation resulting in considerable variation in relation to the level of integration (Cosgrove et al, 2022; Marcus-Quinn et al., 2019). As more schools have adopted one-to-one digital device initiatives, technology is being used more frequently, however the extent to which it is being used to support more student-led practices is largely unknown as there is an absence of large-scale research studies exploring the ways in which this technology is being used within the classroom setting across the curriculum. That being said, there is evidence that schools are partnering with technology companies and technology vendors to adopt specific operating systems and their associated software products on a school wide basis (McGarr, 2026). It is within this climate of techno positivity that the Framework for Junior Cycle and CBAs were introduced.

In order to explore the extent to which digital technologies were used to support students' presentations of their classroom-based assessments (CBAs), a review of the existing literature was undertaken.

Approach guiding the literature review

Guided by the key research question, 'what evidence exists in the literature that schools have become locked-in to using slideware technologies in enacting the CBAs?', a conceptual review of the existing literature was undertaken. This conceptual review was undertaken as sources were too sparse to justify a systematic review. Two criterion from Hulland (2020) were used to guide this conceptual review, process and value:

1. In relation to process, we identified and included all relevant papers related to the roll out of the Junior Cycle Framework (Thorough); identified key themes, and did not overly detail the description of the source materials (Focused);
2. In relation to value, we emphasised new elements emerging from the review, made connections with other disciplinary areas (Novel/Engaging), and made some suggestions for future research (Provide direction).

Searches were undertaken through the Scopus database and Google Scholar. As there were limited studies in this area, further searches were conducted exploring research on the enactment of Classroom-based Assessments (CBAs). Other forms of 'grey literature' were also accessed including early enactment

reviews of some Junior Cycle subjects. These are reports capturing the early rollout of new subjects to gain an insight into the success of their initial implementation and can often shed light on emerging practices. Relevant policy documents, inspectorate publications and professional association outputs were also searched.

In order to validate some of the claims being made, wider searches on the use of digital technologies in teaching and learning in Irish second-level schools was also undertaken. These auxiliary sources were searched to explore whether they provided details on the types of technologies used and their purposes in post-primary classrooms. This diverse range of sources helped paint a broader picture providing confidence in the claims being made.

There was a relatively small number of studies specifically reporting on the rollout and enactment of the Framework for Junior Cycle. While there were a significant number of sources providing commentary on the nature of the curriculum changes and the wider public debate around their merits or otherwise of the changes, research specifically into classroom practices was limited. Furthermore, while there exists a significant amount of pilot studies and experimental work on the use of different types of digital technologies in Irish second-level classrooms, and specifically within key subject areas, there are few studies that paint a broad overall picture of the nature of digital technology use in schools. That being said, some key studies have helped provide an overall picture that can be reported on with some degree of confidence. In total, two Department of Education publications (DES, 2008a, 2015), one NCCA report (NCCA, 2020), six academic papers (Carroll et al. 2024; Cosgrove et al 2014; Dunne et al. 2020; Johnston et al 2015; McCoy et al 2016; Scully et al. 2025) and 3 reports from a longitudinal study on the Junior Cycle (McGarr et al, 2022, 2023, 2024 were included in the review, as well as a search of school and educational websites. We now provide evidence from this review of the literature that digital technologies have played a key role in the enactment of CBAs. After this, we consider what gets locked-in as a result of CBAs being understood and enacted in this way.

The role of digital technologies in supporting the rollout of CBAs

The CBA is described in the curriculum policy document, A Framework for Junior Cycle, as encompassing *'a broad range of activities including oral presentations, written work of different types, practical or designing and making activities, artistic performances, scientific experiments, projects or other suitable tasks'* (DES, 2015, p. 37). However, there are multiple sources within the literature that suggest that digital technologies play a critical role in the realisation of CBAs in many contexts. While there is no specific reference to the use of digital technologies in the associated curriculum guidelines, a number of studies make reference to the use and reliance on slideware technology (normally MS PowerPoint) to assist students in presenting their work. A four-year longitudinal study exploring schools' enactment of these curriculum changes (McGarr et al, 2022, 2023, 2024) found that one of the major challenges to realising the reforms was access to digital technologies for students to undertake project-based learning associated with the CBAs. Across this longitudinal study, both teachers and school principals noted challenges in gaining access to schools' computer rooms or laptops so that students could utilise the various technologies as part of this project work. School contexts that had such access, spoke about the difference and ease of enactment created by digital technologies. This is reflected in the following quotes from teachers:

> There is a trolley full of laptops that you can bring to the classroom if you are doing a CBA (McGarr et al. 2023, p. 69)

> From day 1 in first year students are able to work with digital media and are very creative as well, creating presentations, doing projects … from first years and right up … without the Chrome books, I don't think we would have as much success in the CBAs (McGarr et al. 2023, p. 70)

In comparison, schools with limited access and/or digital resources found themselves struggling to enact the CBAs effectively. This study provided an example of a school who did not initially have access to such digital resources but recently purchased laptops. This was perceived as a game changer in enacting the CBAs:

> It's amazing, because you can go on and book it for whatever time period you want … when it comes to CBAs, because you might have 2 CBAs going on at the one-time … a lot of the kids may not have access [to

> technology] at home. Being able to provide that for them … It was a massive addition to the school (McGarr et al. 2023, p. 70)

While the scale of these challenges varied depending on the level of digital technology and digital resources available in school, it does point out the significant role technology has played in realising project-based learning. This longitudinal research study also reported that the use of slideware technologies was the dominant way in which many students presented their CBA work to their peers and teachers, with students explaining how:

> I think CBAs are way too boring. They're all the same. Plain and boring. Always a PowerPoint and you always just search the information online, copy and paste. They could definitely be more fun. If we had more choice, we might do something that you're actually interested in (McGarr et al. 2023, p. 134)

> So having your own choice of like how to represent what you've learned. You can do Like a PowerPoint of stuff. That's what I found more exciting about it because you can represent it without the stress and it's kind of easier to do and easier to like actually to remember what you were doing (McGarr et al. 2023, p. 133)

Other recently published research mirrors the findings of McGarr et al (2023). Work by Carroll et al. (2024), specifically focusing on voluntary secondary schools, schools publicly funded but privately owned and managed, often by religious organisations, found a strong reliance on PowerPoint within teaching and learning, including CBAs. One participating student, for example, explained how they mainly use technology when completing CBAs, for '*Googling, PowerPoint, Kahoot*' (p. 149). While participating students were largely positive about the use of technology within teaching and learning, some expressed concerns regarding the excessive use of slideshow technology (ibid, p. 145).

A further study by Scully et al. (2025) described the experiences of a teacher who teaches '*in a school with very little ICT so we weren't able to give the students the resources they should have had to do their CBAs' (p. 87)*. Further reference to slideware technologies, specifically PowerPoint, and their role in supporting students' presentation were also noted in the early enactment review of the English curriculum (NCCA, 2020). Extensive use of PowerPoint was also reported in a study by McCoy et al (2016) exploring teaching and learning in second level schools and the rollout of high-speed broadband. While not explicitly looking at CBAs, work by Dunne et al. (2020), also found an overreliance on technology in teaching and learning, which was often unplanned and lacked a clear purpose. This evidence is supported by other studies into the use of digital technologies in Irish classrooms going back several years indicating a high level of use of slideware technology in supporting presentations in classrooms. Research by Johnston et al (2015) into enacting a skills-based curricula in schools reported that many students used PowerPoint to share their learning with peers but that the software was limited in its capacity to reflect the full extent of pupil learning. Cosgrove et al (2014) also found high levels of use of slideware technologies by teachers. Teachers reported most frequently using applications such as word processing and presentation software to prepare resources for lessons. A similar finding by the inspectorate into the reliance on slideware technology was also reported as far back as 2008 (DES, 2008a).

Therefore, while not specifically referenced within the curriculum guidelines, the use of slideware technology appears to have become the default way in which many students present their project-based learning. Further evidence of its dominance is reflected in some websites that provide informal guidelines for students and teachers. They specifically reference the number of slides required for each project presentation. Some school websites, for example, have issued guidelines around how many slides are needed for CBA presentations and what such slides should contain in terms of the number of images and references. For example, an information website from one second-level school advising students about their CBA in English advises that:

> Once you have your information you should make a PowerPoint presentation to help you get your point across by giving your audience something to read. … The layout of the presentation should have all the relevant information in short 1-3 line bullet points. You should have about 4 bullet points per slide. You must add lots of images in your PowerPoint. This gives your audience something to look at, so they don't get bored when you're talking. (School website, 2018)

There also exist several commercial websites providing advice to students which offer similar types of guidance:

> A student, pair or group can choose a topic or area of interest and present on the topic. Students may want to use media or props to elevate their discussion, and the presentation will then be followed by questions from the teacher. Students may want to create a title page for their CBA 1 English PowerPoint so that their work is clearly formatted and well structured. (https://www.twinkl.ie/blog/5-fantastic-cba-1-english-ideas)

Drawing on this range of diverse sources, this literature review indicates that slideware technologies have been extensively used in Irish classrooms for nearly two decades and that more recently they appear to play a particularly important role in facilitating the presentation of students' project-based learning as part of the curriculum changes. This indicates that this type of slideware software has become a common mechanism through which students present their work. In this way, the slideware technologies become the main conduit through which students experience the CBA. We now consider the consequences of this by considering what gets locked-in as a result of this reliance on slideware technology.

What gets locked in through reliance on slideware technology?

While acknowledging that such software provides opportunities for students to collate and present their ideas to their peers, the dominance of such software has not been subject to any critique within the Irish context to date, despite its monopoly in classrooms. This section considers what is locked-in at a classroom level when slideware technologies become the predominant way in which student work, in this instance their CBAs, are presented as evidence of project work. Acknowledging there are many different consequences to the reliance on slideware technologies, we offer two potential consequences of this reliance on slideware technologies for the realisation of the CBAs: locking-in to linear ways of thinking and locking-in to a focus on skills development for future work.

Locking-in to linear ways of thinking

Gries and Brooke (2010) argue that Microsoft PowerPoint, the most widely recognised slideware software, has achieved such a level of market saturation that its name has become a stand-in term for all slideware technologies. Such is the dominance of the software as a tool to support presentations, it has become synonymous with the activity itself. 'Giving your PowerPoint presentation' has come to describe the act of presenting so much so that completing the slides of a planned presentation is seen as sufficient planning for the task itself. Like many products with such a level of market dominance, its use often goes unquestioned. However, as Khoo et al. (2014) advise;

> Software is not neutral. It comes with social and cultural assumptions that afford particular actions while constraining others. Software is designed for particular purposes and embeds particular values in its affordances such that particular ways of thinking and acting are more likely than others. (Khoo et al., 2014, p. 31)

Microsoft PowerPoint, the first significant slideware technology to become established in education, was not originally designed for education. This product was originally designed as a business/marketing presentation tool. Ever since their introduction into education through their inclusion with the Microsoft suite of office applications, their suitability as a teaching tool has been the subject of much debate (Duffy & Henry, 2016). Debates about the educational effectiveness of PowerPoint have tended to look at it from two different perspectives. There is a considerable body of research literature exploring educational effectiveness of PowerPoint as a presentation tool comparing it to more traditional forms of teacher presentation. This body of research has indicated that the use of slideware technologies had no significant effect on learning (Baker et al, 2018). There is another body of research and commentary raising questions around the educational value of slideware technologies from a more cognitive perspective (Garner & Alley, 2016; Paul & Cicek, 2021; Tufte, 2003). This body of work has tended to critique its use in education on the basis that the way PowerPoint structures information can significantly influence students' underlying internal knowledge structure (Duffy & Henry, 2016). These criticisms arise from the linear format in which slideware technologies such as PowerPoint present information. This linearity is also extended to the content of each individual slide where default slide layouts prioritise the use of bullet

points to express ideas and information. The presentation of deep and interconnected knowledge in this linear way has the capacity to fragment knowledge and promote a type of list-like thinking that does not reflect the complex concepts that the student may be exploring as part of their project work. For example, Adams (2006) contends that the software and its style of presentation of information, '*supports a cognitive style inconsistent with the development of higher analytical thinking skills*' (Adams, 2006, p. 390). Adams further explains that;

> PowerPoint's decidedly linear slide-sequence is both a strength and a weakness. The pre-determined deck helps map out a clear, singular course for both teachers and students to follow. It is efficient, expedient. However, this one-way-ness can also render a set of PowerPoint slides less valuable pedagogically if the students' learning ends up being forced mechanically along an inappropriate path. (Adams, 2006, p. 404)

Garner and Alley (2016) offer as similar critique of the software arguing that the default setting of the software prioritises the use of text as the main mode of presentation. While they acknowledge that images and other media can be used, they are, '*offered as a secondary choice to the bulleted list*' and '*simply add situational interest or visual entertainment, thus representing a harmful disconnection between the graphic and instructional purpose of the slide*' (p. 1567). These limitations in the software have led to calls for more freeform types of presentation software to enable teachers and learners to present information in an organic and non-linear format with less dependency on text, enabling interconnectedness of concepts and ideas to be captured (Tufte, 2003).

Looking at the dominant use of slideware technologies to present student project work at Junior Cycle, it could be argued that students are locked-in to presenting their work in this linear manner. Therefore, at a learning level, students are confined to presenting their insights in a particular way. As a result, they are also locked out of presenting their learnings in other formats that more accurately reflect the complex and interconnected nature of the ideas they are grappling with.

Locking into ways of working

Internationally, many education systems have shifted from a focus on knowledge and content to a focus on skills (Sahlberg, 2016; Verger et al., 2016; Young, 2007). This 'global curriculum turn' (Priestley & Sinnema, 2014) is evident in the fact that there are few examples of education systems that don't place some priority on skills (Priestley et al. 2023, p. 17). The terminology may differ, with jurisdictions adopting differing terms such as Key Skills, 21st Century Skills, Graduate Attributes or Competencies. Such a focus is now so common that Priestley et al. (2023) suggests that the term 21st Century skills has become part of the '*shared supranational discourse*' *(p. 47) and has achieved 'near-global acceptance*' (p. 31).

A focus on skills is informed by concerns regarding students' preparedness for their future lives and careers, ensuring students are equipped with the necessary skills for an uncertain future and workplace (Meeder & Pawlowski, 2020; Levy & Murnane, 2004, 2006; Moltó Egea, 2014). Within such a context, a focus on skills is viewed as the '*Promised Land*' (Papastephanou & Angeli, 2007, p. 604). Globalisation, economic development, and national competitiveness (Yates & Young, 2010) are influential factors here, evident in the important role played by the OECD and other economic policy organisations in this regard (Binkley et al., 2011; Priestley et al. 2023; Voogt & Roblin, 2012; Yates & Young, 2010). As the '*need for service and knowledge workers has grown*' (Voogt et al. 2013, p. 403), so has the focus on skills within education systems. Reflecting a social efficiency ideology, within such contexts the purpose of school is seen as being to;

> … efficiently meet the needs of society by training youth to function as future mature contributing members of society. Their goal is to train youth in the skills and procedures they will need in the workplace and at home to live productive lives and perpetuate the functioning of society (Schiro, 2008, p. 4)

Digital literacy is often presented as a core skill students need to be successful within a changing and dynamic world (Daggett, 2010; Lamb et al. 2017; Marcus-Quinn et al., 2019; OECD, 2005). Others, however, have challenged these assumptions, questioning whether a focus on skills will enable students to become the type of citizens and workers such policies seek, questioning whether these skills 'may be

redundant by the time children enter the workforce'. (Priestley et al. 2023, p. 19). The development of ever-more sophisticated generative AI systems is adding further weight to the redundancy argument.

The use of digital technologies in the curriculum reforms at Junior Cycle need to be seen within this wider context. The use of slideware technologies for the presentation of student project work is not only used because of its availability, it is also used in the belief it is contributing to the development of important digital skills for the students' future. A similar focus is evident in other countries, for example, Lake (2022) explored how cloud-based office productivity suite(s) such as Google Workspace have been used to engage students in their learning while also preparing them for the workplace. Examples included how students used the tools to provide feedback; complete a SWOT analysis of a business; use Google Forms with mobile phones to record observations of the emotional state of individuals; and create a sustainability presentation of some of these processes. It is argued by others that not only do such practices aim to prepare students with the digital skills for the future workplace, they also inculcate the student to a corporate culture with its related neoliberal and consumerist ideologies (Marone & Heinsfield, 2023; Nixon et al., 2018; Perry et al., 2026).

This form of neo-vocationalism (Benson, 1997; Neuhouser et al., 2023), focusing on a narrow and arguably flawed assumption that proficiency in using slideware technologies directly contributes to workplace readiness, prioritises the interests of potential employers (i.e. big EdTech corporations) in shaping the pedagogical experience, while students' interests are marginalised (Bradford, 2003). This narrow instrumental perspective of education has been subject to critique. Biesta's (2015, 2020, 2022) concern around the shift from education to learnification of education captures these concerns where he argued that education (and schools) 'can never just be a matter of adaptation to the existing societal order but need to have an orientation towards their independence' (p. 326). Within a techno-positive climate, where all digital practices are seen as progressive, the over reliance on technologies such as slideware presentations may instead disempower learners, providing limited pedagogical opportunities to engage in deep learning. It reflects an emphasis on completing the digital artefact (the product), rather than the educational journey one has travelled acquiring understanding and presenting the knowledge they have gained (the process). In the case of the CBA, rather than focusing on the learning outcomes associated with the project, becomes conceptualised as a task that involves the completion of a set of slides to be ultimately presented or submitted to their teacher. What is locked-in in this instance is an understanding of what the project work entails and its purpose. This potentially complex educational project has arguably become reduced to the completion of a digital artefact and the backwash effect on the nature of the pedagogy used to achieve this can be significant.

Conclusion: locked-in and locked-out?

Cone and Lai's (2025) exploration of how schools can become dependent on particular software providers and how schools can become locked-in through this dependency acted as the starting point for this exploration. Through a review of the literature, the study has highlighted how this lock-in at an institutional level can percolate down to the grassroots level of the classroom influencing how project work is conceptualised and how curriculum changes, in this case the CBAs, are enacted. The dominance of the use of slideware technologies within schools has resulted in these technologies becoming the default way in which much of the student work in their CBAs (project work) is compiled and presented. We have provided two potential examples of what is locked-in within this instance, locking-in to ways of thinking and ways of working. The insights from this work have implications for curriculum enactment. The availability of digital tools and their dominance in schools adds a further level of complexity to curriculum enactment as the presence of such tools can influence how curriculum reforms are enacted. While they can facilitate changes, they can also inhibit the realisation of other aspects. The use of slideware technology is a particularly good example. While they provide a platform to enable learners to compile their thoughts, present their work to their peers and teacher using various media, a reliance on this software can also limit other forms of expression and straight-jacket students' presentations into a linear mode that may not reflect the complexity and depth of the student's insights.

While this paper has explored the various ways in which lock-in influences the nature of teaching and learning at the classroom level, before concluding it is worth considering what is also locked-out

through this process. Eisner's (1985) concept of the null curriculum is useful in this regard. The null curriculum refers to what is not taught in schools either deliberately or accidentally. At the surface level it could be argued that restricting students' exposure to specific technologies and operating systems narrows the spectrum of digital tools they have exposure to. While this is a valid argument, this technology-focused perspective fails to consider the other opportunities that are excluded from their field of experiences through the emphasis on the use of this form of technology. For example, using slideware technologies to present insights and learnings from their project work (rather than more traditional paper-based, arts-based or other media formats) may limit their abilities to present the complexities of what they have learned and limits opportunities for creativity, particularly if they are forced to reduce their insights into linear bullet point slides. Restricting their evidence of learning to digital artefacts also narrows the field of opportunities for them to evidence and celebrate their learning in other ways. Our increasing familiarity with slideware presentation formats also limits our abilities to consider alternative ways of working and presenting information. Therefore, while it is important to acknowledge the ways in which schools are locked-in to particular technologies through their dependence on specific platforms and technology providers, it is equally important to consider what is also locked-out through this process. As the digital transformation of education continues and as further technologies become embedded as normal practice, educators need to critically reflect not only on the affordances of these technologies but also what is lost, and locked-out, through this process of change. This paper has explored just one taken-for-granted piece of software commonly used in schools and has raised questions in relation to its educational merits and the ethics of its dominance in schools. Similar questions could be asked in relation to a multitude of other digital technologies and software products within schools that are used on a daily basis and are not subject to any scrutiny.

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References

Adams, C. (2006). PowerPoint, habits of mind, and classroom culture. *Journal of Curriculum Studies*, *38*(4), 389–411. https://doi.org/10.1080/00220270600579141

Baker, J. P., Goodboy, A. K., Bowman, N. D., & Wright, A. A. (2018). Does teaching with PowerPoint increase students' learning? A meta-analysis. *Computers & Education*, *126*, 376–387. https://doi.org/10.1016/j.compedu.2018.08.003

Benson, C. S. (1997). New vocationalism in the United States: Potential problems and outlook. *Economics of Education Review*, *16*(3), 201–212. https://doi.org/10.1016/S0272-7757(96)00073-8

Biesta, G. J. J. (2015). What is education for? On good education, teacher judgement, and educational professionalism. *European Journal of Education*, *50*(1), 75–87. https://doi.org/10.1111/ejed.12109

Biesta, G. J. J. (2020). What constitutes the good of education? Reflections on the possibility of educational critique. *Educational Philosophy and Theory*, *52*(10), 1023–1027. https://doi.org/10.1080/00131857.2020.1723468

Biesta, G. J. J. (2022). School-as-institution or school-as-instrument? How to overcome instrumentalism without giving up on democracy. *Educational Theory*, *72*(3), 319–331. https://doi.org/10.1111/edth.12533

Binkley, M., Erstad, O., Herman, J., Raizen, S., Ripley, M., Miller-Ricci, M., & Rumble, M. (2011). Defining twenty-first century skills. In *Assessment and teaching of 21st century skills* (pp. 17–66). Springer Netherlands.

Borges, J., McInerney, C., Exton, C., Harkin, B., & McGarr, O. (2024). The subjective nature of curriculum innovation: The case of computer science students in upper secondary education in Ireland. *Computer Science Education*, 5(4), 820–848.

Bradford, M. (2003). *"Learning to sell yourself": A qualitative critique of neo-vocationalism as a school-business partnership* [Doctoral dissertation, University of Canterbury]. https://ir.canterbury.ac.nz/items/ca9bb70f-38c2-4c5e-8f0a-f8f70673e45d

Carroll, E., MacCoy, S., & Ye, K. (2024). *"Embracing diversity in all its forms": The voluntary secondary sector in Irish education (No. 182)*. Research Series.

Cone, L., & Lai, S. S. (2025). Infrastructural dependency in the datafied welfare state: The case of Google Chromebooks. *Learning, Media and Technology*, *50*(1), 44–60. https://doi.org/10.1080/17439884.2024.2437680

Cosgrove, J., Butler, D., Leahy, M., Shiel, G., Kavanagh, L., & Creaven, A. M. (2014). *The 2013 ICT census in schools–main report*. Educational Research Centre. https://www. erc.ie/documents/ict_census2013_summaryreport. pdf

Cosgrove, J., Feerick, E., Moran, E., & Perkins, R. (2022). *Digital technologies in education – Ireland in the international context: Trends and implications from PISA*. Educational Research Centre.

Daggett, W. R. (2010). Preparing students for their technological future. *International Center for Leadership in Education*, *1*, 14.

Department of Education and Science. (1997). *Schools IT2000—A policy framework for the new millennium*. Stationery Office.

Department of Education and Science. (2000). *Blueprint for the future of ICT in Irish education: Three year strategic action plan 2000-2003*. Stationery Office.

Department of Education and Science. (2008a). *Investing effectively in information and communication technologies in schools 2008-2013—The report of the minister's strategy group*. Stationery Office.

Department of Education and Science. (2008b). *ICT in schools: Inspectorate evaluation studies*.

Department of Education and Science. (2009). *Smart schools = Smart economy: Report of the ICT in schools joint advisory group to the minister for education and science*. Stationery Office.

Department of Education and Science. (2015). *Digital strategy for schools 2015-2020: Enhancing teaching, learning and assessment*. Stationery Office.

Department of Education and Science. (2017). *Digital strategy for schools: Action plan 2015-2020*. Stationery Office.

Department of Education and Skills. (2015). *A framework for junior cycle 2015*.

Department of Education. (2022). *Digital strategy for schools to 2027*. Stationery Office. https://assets.gov.ie/static/documents/digital-strategy-for-schools-to-2027.pdf

Duffy, R. M., & Henry, M. (2016). Death by powerpoint: Alternatives for life. *BJPsych Advances*, *22*(6), 420–427. https://doi.org/10.1192/apt.bp.114.013078

Dunne, C., Marcus-Quinn, A., & Ó Dalaigh, C. (2020). *Report of the Independent Review Group (IRG) on the use of tablet devices in Ratoath College (RC)*. https://hdl.handle.net/10344/9686

Eisner, E. W. (1985). *The educational imagination: On the design and evaluation of school programs* (2nd ed.). Macmillan Publishing Co.

Garner, J. K., & Alley, M. P. (2016). Slide structure can influence the presenter's understanding of the presentation's content. *International Journal of Engineering Education*, *32*(1), 39–54.

Gries, L. E., & Brooke, C. G. (2010). An inconvenient tool: Rethinking the role of slideware in the writing classroom. *Composition Studies*, *38*(1), 11–28.

Hulland, J. (2020). Conceptual review papers: Revisiting existing research to develop and refine theory. *AMS Review*, *10*(1–2), 27–35. https://doi.org/10.1007/s13162-020-00168-7

Johnston, K., Conneely, C., Murchan, D., & Tangney, B. (2015). Enacting key skills-based curricula in secondary education: Lessons from a technology-mediated, group-based learning initiative. *Technology, Pedagogy and Education*, *24*(4), 423–442. https://doi.org/10.1080/1475939X.2014.890641

Khoo, E., Hight, C., Cowie, B., Torrens, R., & Ferrarelli, L. (2014). Software literacy and student learning in the tertiary environment: PowerPoint and beyond. *Journal of Open, Flexible and Distance Learning*, *18*(1), 30–45. https://doi.org/10.61468/jofdl.v18i1.217

Lake, J. M. (2022). Using cloud-based collaborative office productivity tools (Google Workspace) to engage students in their learning and prepare them for the workplace. In E. Sengupta & P. Blessinger (Eds.), *ICT and innovation in teaching learning methods in higher education* (pp. 161–179). International Higher Education Teaching and Learning Association. https://doi.org/10.1108/S2055-364120220000045009

Lamb, S., Maire, Q., & Doecke, E. (2017). Key skills for the 21st century: An evidence-based review.

Levy, F., & Murnane, R. J. (2004). *The new division of labor: How computers are creating the next job market*. Princeton University Press.

Levy, F., & Murnane, R. J. (2006). Why the changing American economy calls for twenty-first century learning: Answers to educators' questions. *New Directions for Youth Development*, *2006*(110), 53–62. https://doi.org/10.1002/yd.167

MacPhail, A., Halbert, J., & O'Neill, H. (2018). The development of assessment policy in Ireland: A story of junior cycle reform. *Assessment in Education: Principles, Policy & Practice*, *25*(3), 310–326. https://doi.org/10.1080/0969594X.2018.1441125

Marcus-Quinn, A., Hourigan, T., & McCoy, S. (2019). The digital learning movement: How should Irish schools respond? *The Economic and Social Review*, *50*(4), 767–783.

Marone, V., & Heinsfield, B., D. (2023). Everyone pursuing their dreams": Google's and Microsoft's discourse on educational technology. *Computers and Education Open*, *4*, 100138. https://doi.org/10.1016/j.caeo.2023.100138

McCoy, S., Lyons, S., Coyne, B., & Darmody, M. (2016). *Teaching and learning in second-level schools at the advent of high-speed broadband*. ESRI.

McGarr, O. (2023). Investigating student teachers' responses to including sustainable development as part of their professional digital competence development. *Technology, Pedagogy and Education*, *32*(5), 653–665. https://doi.org/10.1080/1475939X.2023.2265934

McGarr, O. (2026). The corporate branding of schools in Ireland: Why the absence of a critical debate? *Irish Educational Studies*, 1–15. https://doi.org/10.1080/03323315.2026.2614090

McGarr, O., & Johnston, K. (2021). Exploring the evolution of educational technology policy in Ireland: From catching-up to pedagogical maturity. *Educational Policy*, *35*(6), 841–865. https://doi.org/10.1177/0895904819843597

McGarr, O., McCormack, O., O'Reilly, J., Hennessy, J., O'Meara, N., Calderón, A., Leahy, K., MacPhail, A., Lenihan, R., Neary, A., Lynch, R., Power, J., Ó Gallchóir, C., McMahon, J., & Goos, M. (2024). *Exploring the introduction of the Framework for Junior Cycle – A longitudinal study: Interim Report No. 4 (School Case Studies – Round 2)*. University of Limerick. https://doi.org/10.34961/researchrepository-ul.27443502

McGarr, O., McCormack, O., O'Reilly, J., Hennessy, J., O'Meara, N., Neary, A., Lynch, R., Power, J., Lenihan, R., Calderon, A., MacPhail, A., Leahy, K., Gallchóir, C., McMahon, J., Ní Chathasaigh, C., & Goos, M. (2023). *Exploring the introduction of the Framework for Junior Cycle – A longitudinal study: Interim report no. 2*. University of Limerick. https://researchrepository.ul.ie/articles/report/Exploring_the_introduction_of_the_Framework_for_Junior_Cycle_A_longitudinal_study_Interim_report_No_2/22656892/1?file=40506527

McGarr, O., McCormack, O., O'Reilly, J., Lynch, R., Power, J., Ó Gallchóir, C., McMahon, J., Hennessy, J., Leahy, K., O'Meara, N., Calderon, A., MacPhail, A., Neary, A., Ni Chatasaigh, C., Costello, H., & Goos, M. (2022). *Exploring the introduction of the Framework for Junior Cycle – A longitudinal study: Introductory report (Initial perspectives on implementation, outcomes, Framework for Junior Cycle research study – Final Report and impact)*. University of Limerick. https://researchrepository.ul.ie/articles/report/Exploring_the_introduction_of_the_Framework_for_Junior_Cycle_A_longitudinal_study_Introductory_report_Initial_perspectives_on_implementation_outcomes_and_impact_/19819909?file=38006391

Meeder, H., & Pawlowski, B. (2020). *Preparing our students for the real world: The education shift our children and future demand*. National Center for College and Career Transitions. https://www. nc3t. com/wp-content/uploads/2020/02/Preparing-Our-Students-for-the-Real-World-021720. pdf

Moltó Egea, O. (2014). Neoliberalism, education and the integration of ICT in schools. A critical reading. *Technology, Pedagogy and Education*, *23*(2), 267–283. https://doi.org/10.1080/1475939X.2013.810168

NCCA. (2010). *Innovation and identity: Ideas for a new junior cycle*. https://www.ncca.ie/media/2097/innovation_and_identity_ideas_for_a_new_junior_cycle.pdf

NCCA. (2020). *Report on the enactment of Junior Cycle English 2020*. https://curriculumonline.ie/getmedia/47da7de5-af3f-4ab3-a51d-58c1d4275fc0/jc-english-review-2020_en.pdf

Neuhouser, J. D., Latz, O. A., & Mulvihill, T. M. (2023). Remembering new vocationalism: Reinforcing the future of workforce and professional development in the community college, community college. *Community College Journal of Research and Practice*, *47*(6), 401–412. https://doi.org/10.1080/10668926.2022.2045646

Nixon, E., Scullion, R., & Hearn, R. (2018). Her majesty the student: Marketised higher education and the narcissistic (dis) satisfactions of the student-consumer. *Studies in Higher Education*, *43*(6), 927–943. https://doi.org/10.1080/03075079.2016.1196353

OECD. (2005). *New perspectives on ICT skills and employment*. OECD Digital Economy Papers, No. 96. OECD Publishing. https://doi.org/10.1787/232342747761

Papastephanou, M., & Angeli, C. (2007). Critical thinking beyond skill. *Educational Philosophy and Theory*, *39*(6), 604–621. https://doi.org/10.1111/j.1469-5812.2007.00311.x

Paul, J. W., & Cicek, J. S. (2021). The cognitive science of PowerPoint. *Proceedings of the Canadian Engineering Education Association (CEEA)*, https://doi.org/10.24908/pceea.vi0.14872

Perry, L. B., Yoon, E.-S., Sciffer, M., & Lubienski, C. (2026). The impact of marketization on school segregation and educational equity and effectiveness: Evidence from Australia and Canada. *International Journal of Comparative Sociology*, *67*(1), 175–196. https://doi.org/10.1177/00207152241227810

Priestley, M., & Sinnema, C. (2014). Downgraded curriculum? An analysis of knowledge in new curricula in Scotland and New Zealand. *The Curriculum Journal*, *25*(1), 50–75. https://doi.org/10.1080/09585176.2013.872047

Priestley, M., Angier, C., Schuler, B., & Smith, J. (2023). *Towards a typology of curriculum policy approaches*. UNESCO.

Ryan, B., McGarr, O., & McCormack, O. (2020). Underneath the veneer of techno-positivity–exploring teachers' perspectives on technology use in Further Education and Training. *Teachers and Teaching*, *26*(5–6), 414–427. https://doi.org/10.1080/13540602.2020.1863207

Sahlberg, P. (2016). The global educational reform movement and its impact on schooling. In K. Mundy, A. Green, B. Lingard, & A. Verger (Eds.), *The handbook of global education policy* (pp. 128–144). Wiley Blackwell.

Schiro, M. (2008). *Curriculum theory: Conflicting visions and enduring concerns*. Sage Publications.

School Website. (2018). *2nd year CBA peer advice*. https://hartstowncommunityschool.com/news/2018/2nd-year-cba-peer-advice/

Scully, D., Carroll, M., Clarke, S., & Guirke, G. (2025). Burgeoning teacher assessment identity and learning to 'play the system': The realities of assessment-led reform at lower secondary level in Ireland. *Assessment in Education: Principles, Policy & Practice*, *32*(1), 77–97. https://doi.org/10.1080/0969594X.2025.2467673

Smyth, E., & Hannan, C. (2007). School processes and the transition to higher education. *Oxford Review of Education*, *33*(2), 175–194. https://doi.org/10.1080/03054980701259964

Smyth, E., Dunne, A., McCoy, S., & Darmody, M. (2006). *Pathways through the junior cycle: The experiences of second year students*. The Liffey Press in association with The Economic and Social Research Institute.

Suárez-Guerrero, C., Rivera-Vargas, P., & Raffaghelli, J. (2023). EdTech myths: towards a critical digital educational agenda. *Technology, Pedagogy and Education*, *32*(5), 605–620. https://doi.org/10.1080/1475939X.2023.2240332

Tufte, E. (2003). *The cognitive style of PowerPoint*. Graphics Press.

Verger, A., Fontdevila, C., & Zancajo, A. (2016). *The privatization of education: A political economy of global education reform*. Teachers College Press.

Voogt, J., & Roblin, N. P. (2012). A comparative analysis of international frameworks for 21st century competences: Implications for national curriculum policies. *Journal of Curriculum Studies*, *44*(3), 299–321. https://doi.org/10.1080/00220272.2012.668938

Voogt, J., Erstad, O., Dede, C., & Mishra, P. (2013). Challenges to learning and schooling in the digital networked world of the 21st century. *Journal of Computer Assisted Learning*, *29*(5), 403–413. https://doi.org/10.1111/jcal.12029

Yates, L., & Young, M. (2010). Editorial: Globalization, knowledge and the curriculum. *European Journal of Education*, *45*, 4–10. https://doi.org/10.1111/j.1465-3435.2009.01412.x

Young, M. (2007). *Bringing Knowledge Back In: From social constructivism to social realism in the sociology of education*. Routledge.